

ETHAN GAN

<https://ethangan.netlify.app/>

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SKILLS

Languages: Python, Java, JavaScript, C++, Arduino C, SQL

Frameworks & Technologies: React.js, Node.js, Express, Sequelize, PostgreSQL, HTML, CSS, REST APIs

Tools & Platforms: Git, GitHub, Bash, VS Code, Figma, Anaconda, Azure, SolidWorks, Blender 3D

Certifications: Microsoft Azure AI Fundamentals, Microsoft Azure Fundamentals, TCPS 2: CORE-2022, WHMIS 2015, Machine Shop

Relevant Coursework: Data Structures & Algorithms (C++), Engineering Prototyping (Machining, Sensors, Actuators, Embedded Systems), Visualization (SolidWorks/CAD), Digital Systems, Linear Algebra (MATLAB)

EXPERIENCE

Hardware Team Lead – Quadcopter Drone

Jan 2026 – Present

WATbotics Robotics Student Design Team

Waterloo, ON

- *Previously: Member (Sep 2025 – Dec 2025)*
- Lead hardware development for a quadcopter UAV, spanning from **component planning and procurement** to **hands-on system integration and troubleshooting**.
- Performed conceptual UAV design and flight-performance analysis, evaluating propulsion, battery, and airframe requirements against a target **20+ minute endurance**.
- Diagnose and troubleshoot the UAV propulsion system across the **flight controller, ESCs, motors, and communication signals**, using Mission Planner, a multimeter, and oscilloscope-based measurements.
- Investigate ESC calibration and motor-output issues by testing different control protocols and systematically isolating faults in the flight-control signal chain.
- Led **component planning and procurement**, achieving **20% cost savings** through component selection and purchasing while meeting one-week component lead times.
- Led the **design and planning of the team's new work bay** for UAV development, assembly, and testing.
- Secured **\$1,000+** in funding through successful grant proposal development for the following term.

Policy Innovation Assistant – Transformative Technologies

May 2026 – Aug 2026

CIGI (Center for International Governance and Innovation)

Waterloo, ON

- Conducted a literature review on the emerging field of **Cooperative AI**, synthesizing research on cooperative intelligence, multi-agent systems, evaluation, and governance.
- Developed a draft **theory of change** for CIGI's AI Empowerment Program, using the **Heilmeier Catechism** to structure program objectives, research questions, and proposed activities.
- Built and maintained a structured research base of **100+ sources** using **Zotero and Excel**, organizing literature across research themes and domains.
- Synthesized and presented findings from the **Cooperative AI literature** in a **30-minute research round** for CIGI colleagues.
- Delivered **15-minute weekly research briefings** to the Director of Transformative Technologies, summarizing recent findings and discussing open questions.
- **Awarded Co-op For Good award** by the University of Waterloo.

Firmware and Electrical Team Member

May 2025 – Dec 2025

Midnight Sun Solar Rayce Car Team

Waterloo, ON

- The University of Waterloo's solar-powered endurance race car team competing at the American and World Solar Challenge.
- Implemented a **driver control state machine** in **C** on **FreeRTOS**, wrote **unit tests**, and used **Vagrant** for consistent development.
- Collaborated with various subteams to troubleshoot integration problems.

Engineering Student Experience Analyst

Sep 2025 – Dec 2025

University of Waterloo

Waterloo, ON

- Conducted **end-to-end data analysis** on UWaterloo's well-acclaimed student design teams through a survey and interviews achieving **100% of the targeted population**.
- Built a centralized contact base for student teams through proactive relationship-building and persistent outreach.
- Developed an **Excel/VBA office calendar tool** with **automated monthly layout generation**, centralized event data, and yearly event-cycle views across academic terms.
- Developed **web-scraping tools in Python using BeautifulSoup** to identify wording changes and dead links across **94 webpages**.

- Helped organize a **design sprint** with **Faculty of Engineering leaders and Microsoft** to create an AI-powered tool to improve onboarding time of new university instructors.
- Designed posters and a word-of-mouth strategy for an optional **faculty wide survey** to achieve **10% response rate**.

Project Team Member

Jan 2023 – Apr 2025

WE Accelerate - Microsoft Division

Waterloo, ON

- Led the **product design for a conceptual AI-powered fraud prevention tool** for seniors.
- Collaborated with a 5-person team to create and present a functional design proposal **resulting in the top score and endorsement** as a model presentation for future teams.
- Created **UX flow diagrams and wireframes** to simulate interactions with tools such as **Azure AI Foundry, Azure Search, and DeepSpeech**.
- **Awarded WEA Rippling Award** for standout individual performance during the work term.

Science and Technology Education Instructor

Feb 2023 – May 2025

Principal Academy

Vaughan, ON

- Designed and taught a week-long project-based Summer Coding Camp curriculum for **Python, Java and Scratch** for capacities of **20+ students of grades 4-12**.
- Applied Adobe Creative Cloud to optimize UX, branding, logo design and multimedia content creation and promotional campaigns of **up to 10% conversion rate**.

Coding Contest Creator

Sept 2023 – June 2024

Thornhill Secondary School Computer Club

Thornhill, ON

- Led a team of students to launch coding contests on graph theory and other concepts for **50+ participants**.
- Built **automated input/output test suites** to streamline problem submission evaluation to **lower needed manual verification by 99%**.

PROJECTS

Inventory Management System | React, Node.js, Express, PostgreSQL, SQL, Sequelize ORM, REST APIs

- Built and deployed a **full-stack inventory and procurement system** to centralize equipment tracking, purchase requests, and supplier records for a university design team.
- Designed a **normalized PostgreSQL schema (3NF)** and implemented a **RESTful API** enforcing relational integrity and auditability of inventory flow.
- Owned the system end-to-end, integrating and deploying frontend, backend, and database services using **Netlify, Render, and Supabase**.
- Translated real operational problems into **concrete system requirements**, reducing fragmented inventory ownership across subteams.

Portfolio Website | React, JavaScript, React Router, CSS, HTML, Markdown, Netlify

- Built a responsive, modular **React** website featuring dynamic project showcases with a **Markdown** article system.
- Achieved Lighthouse scores of **92 (Performance)**, **95 (Accessibility)**, and **100 (Best Practices)** through code splitting and semantic HTML.

TankRunners: Procedural Tank Game | Java, JFrame, JPanel, OOP

- Developed a **procedurally generated dungeon adventure game** featuring adaptive AI enemies and dynamic map generation.
- Implemented an AI-driven enemy system utilizing weighted randomness for decision-making and A* pathfinding for optimized navigation.

WalkAlong Hovercraft | Arduino C, SketchUp, TinkerCAD, 3D Printing & Design, Servo/Motors

- Built a functional **1/32 scale hovercraft** to demonstrate airflow and Bernoulli's principle.
- Used Arduino C to communicate rudder movements and motor intensity through a remote receiver, SketchUp for drafting 3D designs, and TinkerCAD for 3D printing.

EDUCATION

University of Waterloo

Expected 2029

Bachelor of Applied Science in Systems Design Engineering, Honours

Waterloo, ON

- Cumulative average: 88%
- **Awards:** President's Scholarship of Distinction (2024), TELUS Entrance Scholarship (2024)